

12. A student was asked to find the identity of a Group 1 metal carbonate by titration.

He was told to use the following method.

- Weigh a sample of the carbonate in a weighing bottle.
- Transfer the carbonate into a beaker and weigh the bottle afterwards.
- Add water to the beaker to dissolve the carbonate.
- Transfer the solution to a volumetric flask.
- Add more water to make the final volume 250.0 cm^3 of solution.
- Accurately transfer 25.0 cm^3 of this solution into a conical flask.
- Add 2–3 drops of a suitable indicator to this solution.
- Fill a burette with 0.100 mol dm^{-3} hydrochloric acid solution.
- Carry out a rough titration of the carbonate solution with the hydrochloric acid.
- Accurately repeat the titration until you get concordant titres and calculate a mean titre.

(a) Another student said that there were two errors in making the 250.0 cm^3 carbonate solution.

Error 1: A small amount of solid remained in the weighing bottle.

Error 2: A small amount of solution remained in the beaker.

Comment on the suggested errors.

If the student is correct suggest how the method could be improved.

If the student is incorrect, explain why.

[2]

Error 1

.....
.....

Error 2

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.....

(b) State why he adds an indicator to this solution.

[1]

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.....

(c) Suggest why he was told to carry out a rough titration first.

[1]

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(d) State what you understand by the term 'concordant titres'.

[1]

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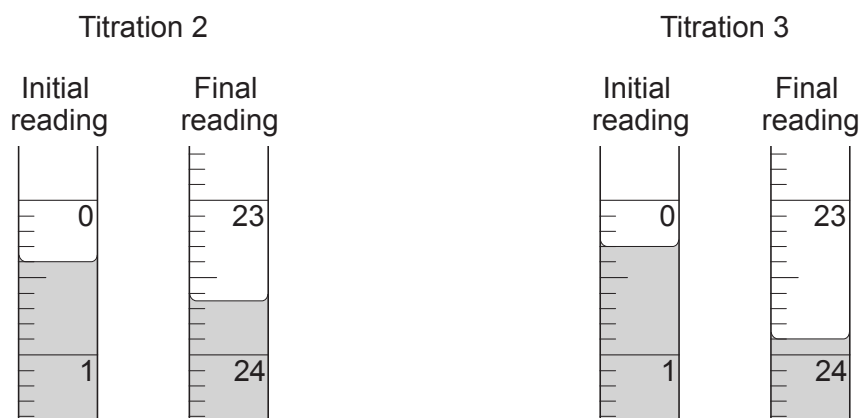
(e) Some of the student's results are shown below:

Mass of weighing bottle + carbonate /g	13.73
Mass of weighing bottle /g	12.48

Titration	Rough	1	2	3
Final reading / cm ³	24.20	23.70		
Initial reading / cm ³	0.00	0.10		
Titre / cm ³				

concentration of hydrochloric acid = 0.100 mol dm⁻³

The diagrams below show the initial burette reading and the final burette reading for the second and third titrations.



Complete the student's table and calculate the mean titre.

[3]

mean titre = cm³



- (f) The equation for the reaction between the metal carbonate and hydrochloric acid is given below. **M** represents the symbol of the Group 1 metal.



- (i) Calculate the number of moles of M_2CO_3 in 25.0 cm^3 of the solution. [2]

number of moles =

- (ii) Calculate the relative formula mass of the carbonate and hence deduce the Group 1 metal in the carbonate.

You **must** show your working. [4]

group 1 metal =

END OF PAPER

